

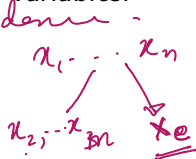
Adding evidence

Given fixed values of a subset of variables $\mathbf{x}_e$ (evidence), find the

- ① Marginal probability queries over a small subset of variables:

- Find $\Pr(\text{Income}=\text{'High'} \mid \text{Degree}=\text{'PhD'})$

$$\Pr(\underline{x_1}) = \sum_{x_2 \dots x_m} \Pr(x_1 \dots x_n \mid \underline{x_e})$$



- ② Most likely labels of remaining variables: (MAP queries)

- Find likely temperature at sensors in a room given readings from a subset of them

$$\mathbf{x}^* = \operatorname{argmax}_{x_1 \dots x_m} \Pr(x_1 \dots x_n \mid \underline{x_e})$$

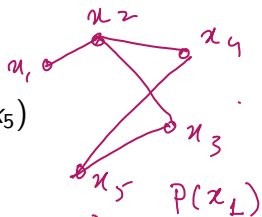
Easy to add evidence, just change the potential.

Adding evidence (Example)

$$P(x_1, x_2, x_3, x_4, x_5) \propto$$

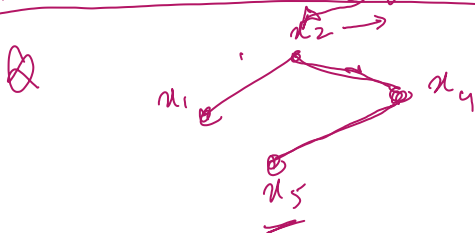
$$\psi_{12}(x_1, x_2) \psi_{24}(x_2, x_4) \psi_{23}(x_2, x_3) \psi_{45}(x_4, x_5) \psi_{35}(x_3, x_5)$$

Compute $P(x_1 | x_3 = 1)$



$$Q(x_1, x_2, x_4, x_5) = P(x_1, x_2, x_4, x_5 | x_3 = 1)$$

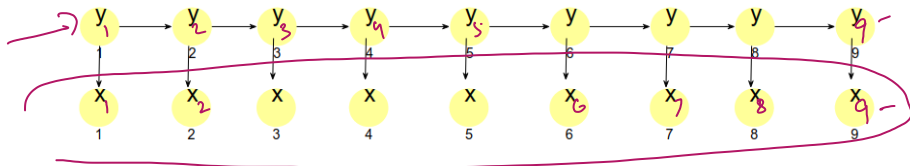
$$\propto \psi_{12}(x_1, x_2) \psi_{24}(x_2, x_4) \underbrace{\psi_{23}^*(x_2, 1)}_{\psi_{23}(x_2, 1)} \psi_{45}(x_4, x_5) \underbrace{\psi_{35}^*(1, x_5)}_{\psi_{35}(1, x_5)}$$



Case study: HMMs for Information Extraction

My review of Fermat's last theorem by S. Singh

t	1	2	3	4	5	6	7	8	9
x	My	review	of	Fermat's	last	theorem	by	S.	Singh
y	Other	Other	Other	Title	Title	Title	other	Author	Author

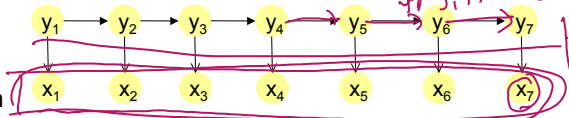


Inference in HMMs

Handwritten notes:

- $\max \Pr(y_7 | \vec{x} = \mathbf{o})$
- $\max_{y_1, y_2, \dots, y_6} \sum \Pr(y_1, \dots, y_7 | x = \mathbf{o})$

- Given,



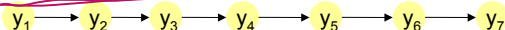
- Graph
- Potentials: $\Pr(y_i | y_{i-1})$, $\Pr(x_i | y_i)$
- Evidence variables: $\mathbf{x} = x_1 \dots x_n = o_1 \dots o_n$.

- Find most likely values of the hidden state variables.

$$\mathbf{y} = y_1 \dots y_n$$

$$\operatorname{argmax}_{\mathbf{y}} \Pr(\mathbf{y} | \mathbf{x} = \mathbf{o})$$

- Define $\psi_i(y_{i-1}, y_i) = \Pr(y_i | y_{i-1}) \Pr(x_i = o_i | y_i)$
- Reduced graph only a single chain of y nodes.

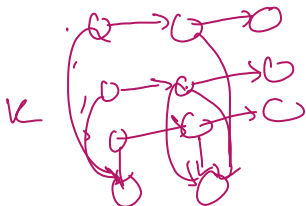


- Algorithm same as earlier, just replace "Sum" with "Max"

This is the well-known Viterbi algorithm

Why approximate inference

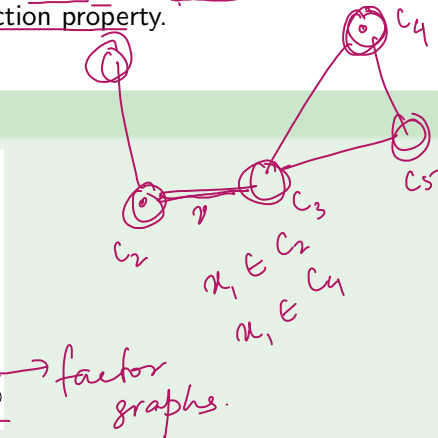
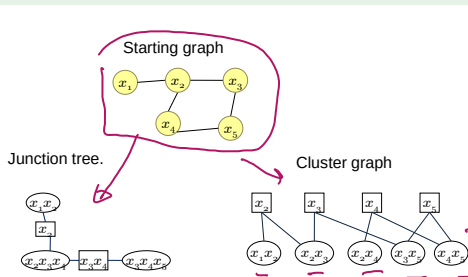
- Exact inference is NP hard. Complexity: $O(m^w)$
 - w = tree width = size of the largest clique in (triangulated) graph-1,
 - m = number of values of each discrete variable in the clique.
- Many real-life graphs produce large cliques on triangulation
 - A $n \times n$ grid has a tree width of n
 - A Kalman filter on K parallel state variables influencing a common observation variable, has a tree width of size $K + 1$



Generalized belief propagation

- Approximate junction tree with a cluster graph where
 - Nodes = arbitrary clusters, not cliques in triangulated graph.
Only ensure all potentials subsumed.
 - Separator nodes on edges = subset of intersecting variables so as to satisfy running intersection property.
- Special case: Factor graphs.

Example cluster graph



Belief propagation in cluster graphs

- Graph can have loops, tree-based two-phase method not applicable.
- Many variants on scheduling order of propagating beliefs.
 - ▶ Simple loopy belief propagation
 - ▶ Tree-reweighted message passing
 - ▶ Residual belief propagation
- Many have no guarantees of convergence. Specific tree-based orders do
- Works well in practice, default method of choice.

For more: Refer text book or read this paper:
K. P. Murphy et al. (1999). Loopy belief propagation for approximate inference: an empirical study. In: Proceedings of the Fifteenth conference on Uncertainty in artificial intelligence. 467475.

Others

- Sampling (to be discussed later)
- Combinatorial algorithms for MAP **boykov01Fast**
- Greedy algorithms: relaxation labeling.
- Variational methods like mean-field and structured mean-field that view inference as an optimization problem.
- LP and QP based approaches.

Learning Graphical Models from Data

- Learning Graph 
- Learning Potentials given fixed graph 

Graph Structure

- ① Manual: Designed by domain expert
 - ▶ Used in applications where dependency structure is well-understood
 - ▶ Example: QMR systems, Kalman filters, Vision (Grids), HMM for speech recognition and IE.
- ② Learned from examples
 - ▶ NP hard to find the optimal structure.
 - ▶ Widely researched, mostly posed as a branch and bound search problem.
 - ▶ Useful in dynamic situations

Recent research in learning Bayesian network structure

- Joint Bayesian Inference of Graphical Structure and Parameters with a Single Generative Flow Network. Tristan Deleu, Mizu Nishikawa-Toomey, Jithendaraa Subramanian, Nikolay Malkin, Laurent Charlin, Yoshua Bengio. NeurIPS 2023
<https://openreview.net/pdf?id=t7lnhhi7De>
- DAGs with NO TEARS: Continuous Optimization for Structure Learning Xun Zheng, Bryon Aragam, Pradeep Ravikumar, Eric P. Xing <https://arxiv.org/abs/1803.01422>

TBD

Parameters in Potentials

① Manual: Provided by domain expert

- ▶ Used in infrequently constructed graphs, example QMR systems
- ▶ Also where potentials are an easy function of the attributes of connected graphs, example: vision networks.

② Learned: from examples

- ▶ More popular since difficult for humans to assign numeric values
- ▶ Many variants of parameterizing potentials.
 - ① Table potentials: each entry a parameter, example, HMMs
 - ② Potentials: combination of shared parameters and data attributes: example, CRFs.

Learning potentials: Two settings

1 Generative:

- 1 $P(\underline{\mathbf{x}}) = P(x_1, \dots, x_n)$ represented as a graphical model $\underline{G}$.
- 2 Samples are $\underline{D} = \{\underline{\mathbf{x}}^1, \dots, \underline{\mathbf{x}}^N\}$
- 3 Potentials to be learned are $\underline{\psi_C(\mathbf{x}_C)}$

2 Conditional:

- 1 $P(\underline{\mathbf{y}}|\underline{\mathbf{x}}) = P(y_1, \dots, y_n|\underline{\mathbf{x}})$ represented as a graphical model $\underline{G}$ over the $\underline{\mathbf{y}}$ variables.
- 2 Training samples are $\underline{D} = \{(\underline{\mathbf{x}}^1, \underline{\mathbf{y}}^1), (\underline{\mathbf{x}}^2, \underline{\mathbf{y}}^2), \dots, (\underline{\mathbf{x}}^N, \underline{\mathbf{y}}^N)\}$
- 3 Potentials to be learned are $\underline{\psi_C(\mathbf{y}_c, \underline{\mathbf{x}})}$

Most of the topics under learning we will discuss apply equally well to the two settings.

Learning potentials

Given sample of data generated from a distribution represented by a graphical model with known structure G , learn potentials.

Two settings: All variables observed or not.

- 1 Fully observed: each training sample has all n variables observed.
- 2 Partially observed: a subset of the variables are observed.

Training follows with classical principle of maximizing likelihood of training data.

Training Bayesian networks (Fully observed case)

$$D \equiv \{(x^1, y^1) - \dots (x^N, y^N)\}$$

$\Theta \equiv$ set of all potentials.

$$y \sim BN$$

$$Y \equiv [y_1, y_2, \dots, y_n]$$

$$LL(\underline{\theta}, D) = \sum_{i=1}^N \log \Pr(\underline{y}^i | \underline{x}^i, \underline{\theta})$$

$$= \sum_{i=1}^N \log \prod_{j=1}^n \Pr(y_j^i | \underline{y}_{Pa(j)}^i, \underline{x}^i, \underline{\theta})$$

$$= \sum_{i=1}^N \sum_{j=1}^n \log \Pr(y_j^i | \underline{y}_{Pa(j)}^i, \underline{x}^i, \underline{\theta})$$

$$= \sum_{j=1}^n \left[\sum_{i=1}^N \log \Pr(y_j^i | \underline{y}_{Pa(j)}^i, \underline{x}^i, \underline{\theta}) \right]$$

Like normal classification task. No challenge arising during training because of graphical model.